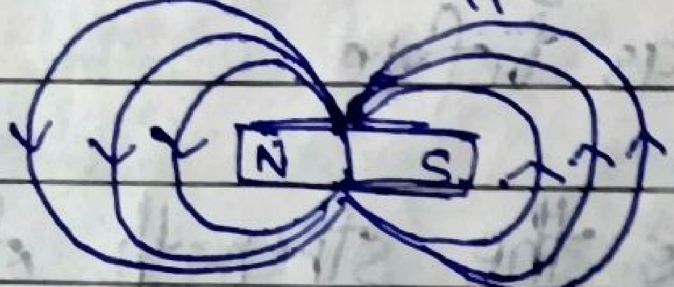
Ch-4 (Notes)

Compass: A compass is a device which has a tiny needle which points towards the North-South & gets deflected by Magnet.

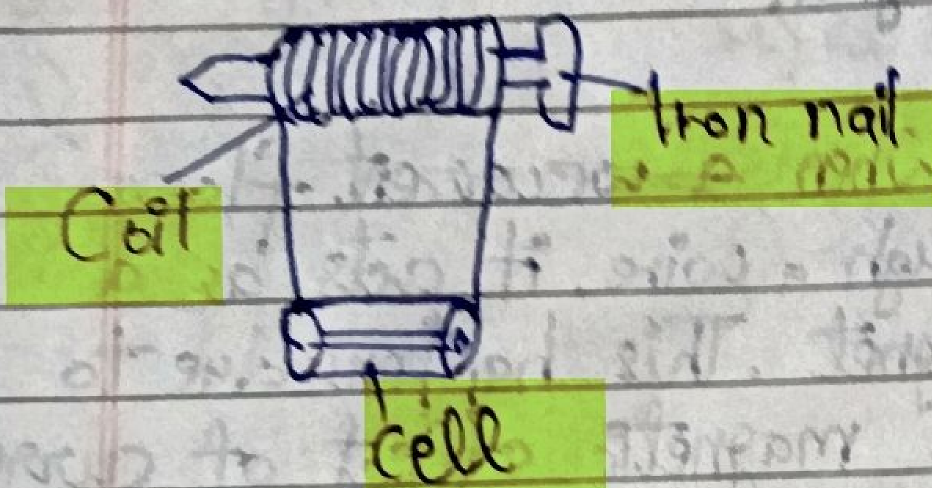
Electromagnets: When ~~a~~ current flows through a wire, it acts as a magnet. This happens due to the magnetic effect of current.

Magnetic field: Region around a magnet (or electromagnet) ~~around~~ where its effects can be felt.



Hans Christian Oersted was a proffesor who taught at a university ~~of~~ in Denmark. He discovered the magnetic effect of current in 1820.

When current passes through a cylindrical coil, it becomes a electromagnet. The core of the electromagnet also increase its ability. Pratically, iron cores are used to make them stronger.



Electromagnets also have a North & South pole. They can also be switched by connecting the wire to opposite terminals as before.

To increase the strength of an electromagnet

- Connecting it a stronger battery
- Increasing the no. of turns in coil

Earth is also a large magnet as inside there are many molten metals, & their movement makes electric currents. These currents make a magnetic field. Many migratory animals use this field to navigate. This field also protect us from harmful particles from the space

Uses

- Cranes
- Junk yards
- Motors
- Generators

When current passes through a conductor, the conductor shows some resistance, nichrome shows high resistance but copper shows less resistance. Due to this resistance, current converts into heat. This is the heating effect of current.

Factors for heating effect of current

- Duration of current flow
- Thickness
- Length
- Material
- Strength of current

Use : Geysers

- Heaters
- Incandescent Bulb
- Iron
- Steel manufacturing industry

Heating can sometime melt plastic parts due to over-plugging, etc & can cause an house-fire. To prevent this, safety devices are installed.

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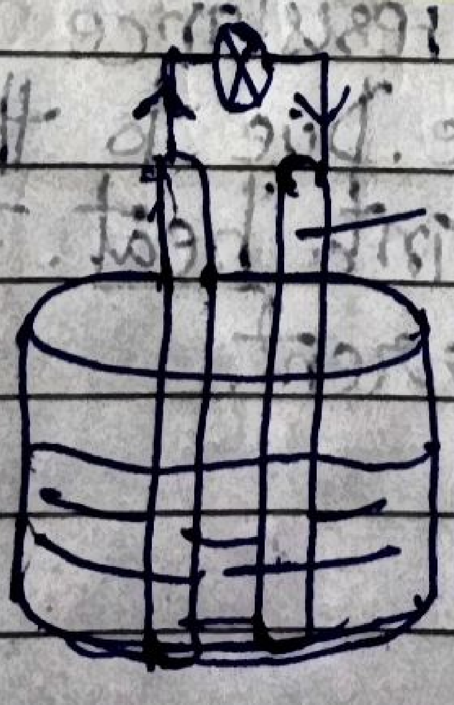


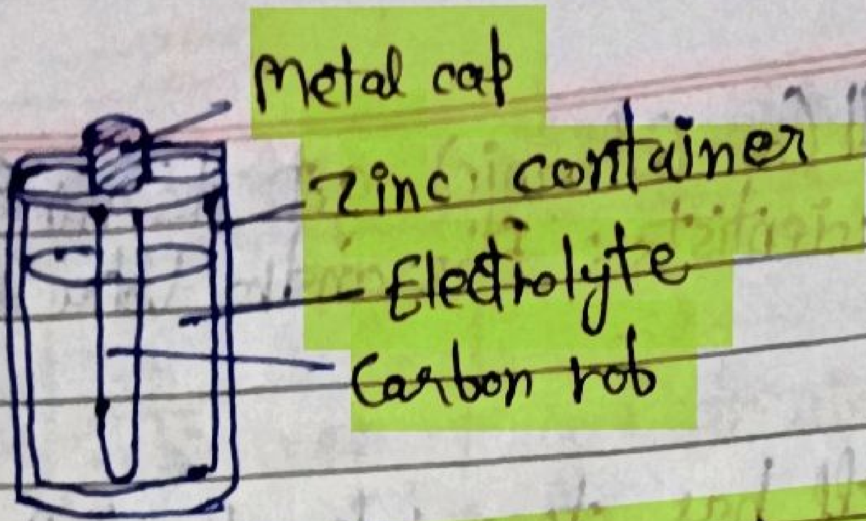
Voltaic cell (or Galvanic) got its name from 2 diffent scientists : Alessandro Volta & Luigi Galvani

A voltaic cell has two rods of different metals called electrodes, which are dipped in a substance called electrolyte.
A chemical reaction b/w metal & electrolyte makes electricity. Over time, this cell's chemicals get used up & cell becomes "dead".

Metal combinations : zinc/copper, zinc/silver copper/iron, magnesium/copper. (Copper is a positive electrode & zinc is a negative)

Dry cells are called 'dry', because in them electrolytes are not liquid but a paste-like substance. They use zinc container as (-) & metal cap as (+). They are non-reusable.

Voltaic cells had a weak acid or salt solution as electrolyte. (~~Th~~ They are non-reusable too)



Rechargable batteries are being used more & promoted as they last longer & are more efficient.

Lithium Iron (Li-ion) battery is becoming more common & can ~~by~~ be found in any device. They rely on lithium ~~&, cobalt~~, cobalt, etc, so countries are trying to secure supply & recycle old batteries.

Solid state battery is a type of battery on which scientists ~~&~~ are working. In this battery, electrolyte, is replaced with a solid substance, which makes the battery last longer. paste